[INCH POUND]
GGG-P-833B
June 20, 1994
SUPERSEDING
GGG-P-833A
January 14, 1966

FEDERAL SPECIFICATION

PUNCHES, CUTTING

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand punches commonly used by saddlers and metalworkers.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type, classes, and styles. The punches shall be of the following types, classes, and styles, as specified (see 6.1):

Type I - Cutting (drive)

Class A - Rectangular (bag)
Class B - Circular

Style 1 - Double bow
Style 2 - Single bow

Class C - Tinners'

Style 1 - Hollow

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Distribution Statement A. Approved for public release; distribution is unlimited.

FSC 5110

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Federal Specification:

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-B-52761 - Belting, Flat: Conveyor, Rubber Industrial Synthetic Reinforcement

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI):

ANSI B209.2 - Punches and Drift Pins, Safety Requirements for Metal

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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American Society for Testing and Materials (ASTM):

ASTM E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of punches otherwise conforming to this specification.

3.2 Material. The material shall be made from quality, fine-grain, hot-rolled or cold-finished carbon or alloy steel bars, or equivalent, conforming to any of the following standards: ASTM A 29/A 29M, ASTM A 322, ASTM A 331, ASTM A 576, or ASTM A 681. All forgings shall be of uniform quality, free from defects, imperfections, flash, and irregular or sharp projections and edges.

3.3 Finish. The punch shall be free from forge flash, oxide scale, rust, fins, sharp edges or corners (except cutting edges), pits, burrs, cracks, and other detrimental defects. Surfaces not usually ground or polished shall have a surface finish conforming to industry practices.

3.4 Identification marking. The punch shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.5 Punch sets. When sets of punches are specified (see 6.1), all punches contained in the set shall conform to the requirements of their respective class and style covered by this specification.

3.5.1 Containers for punch sets. Punches, when ordered in encased sets, shall be furnished in durable wooden or plastic boxes, fabric wraps, plastic wraps, or rolls as specified (see 6.1). The containers shall be provided with a compartment for each individual punch in order to facilitate selection, prevent damage to the punches, and provide a method of storage while the punches are not in use. All containers shall be marked plainly and legibly with the respective set designation.

3.5.1.1 Wooden boxes. The box and cover shall be of hardwood such as maple, ash, hickory, or birch. Unless otherwise specified (see 6.1), the box shall have a natural finish and withstand the drop test specified in 4.4.5.

3.5.1.2 Plastic boxes. The boxes and covers shall be of plastic that will withstand the flammability tests as specified for plastic wraps or rolls in 4.4.4, and not be damaged when subjected to the drop test specified in 4.4.5.

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3.5.1.3 Fabric wraps or rolls. Fabric wraps or rolls shall be durable material such as canvas or duck and shall be provided with durable ties.

3.5.1.4 Plastic wraps or rolls. Plastic wraps or rolls shall be of plastic or plastic-coated cloth. The wraps or rolls shall be provided with snap fasteners or durable ties. The wraps or rolls shall be capable of withstanding the flammability test specified in 4.4.4.

3.6 Type I, cutting (drive) punch. The type I, cutting (drive) punch shall be designed to have a punch end and a properly proportioned shank. The class A and B punches shall be used for punching holes in leather and other similar materials. The class C punch shall be used for punching holes in soft sheet metal. Class A and B punches shall be capable of withstanding the test specified in 4.4.1. The class C punch shall be capable of withstanding the test specified in 4.4.2. For classes A, B, and C punches, the hardness between the cutting edge and a minimum of 1/2 inch from the cutting end shall have a Rockwell C hardness of 45 to 55 or 83.0 to 87.9 on the Rockwell superficial 15N scale when tested as specified in 4.4.3.

3.6.1 Punch end. The class B punch end shall be designed to provide a smooth passage for the punchings through the barrel without difficulty and be discharged automatically. The barrel, adjacent to the cutting edge, shall be smoothly finished and polished.

3.6.1.1 Cutting edge. The cutting edge of the punches shall not dull when subjected to the test specified in 4.4.1 for classes A and B, and 4.4.2 for class C.

3.6.2 Shank. The shank shall be either circular, square, rectangular, hexagonal, or octagonal in shape (taper optional). The striking end face shall be flat or domed, smoothly finished, and at right angles to the longitudinal axis of the punch within 2°.

3.6.3 Class A, rectangular (bag). The class A punch shall have a rectangular shaped cutting edge having semicircular corners, shall be capable of punching holes conforming to the sizes shown in table I, and be similar to figure 1.

TABLE I. Type I, class A, cutting (drive) punch, rectangular (bag)

Size number	Dimension of hole		Length, overall ± 15 percent
	Length (± 0.010)	Width (± 0.010)	
	Inch	Inch	Inches
1	1/4	5/64	4
2	3/8	3/32	4-3/4
3	1/2	7/64	4-13/16

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TABLE I. Type I, class A, cutting (drive) punch, rectangular (bag) (Cont'd)

Size number	Dimension of hole		Length, overall ± 15 percent
	Length (± 0.010)	Width (± 0.010)	
	Inch	Inch	Inches
4	5/8	1/8	4-7/8
5	3/4	9/64	5
6	7/8	5/32	5
7	1	5/32	5
8	1-1/8	11/64	5
9	1-1/4	11/64	5
10	1-3/8	3/16	5
11	1-1/2	13/64	5
12	1-3/4	7/32	5
13	2	1/4	5

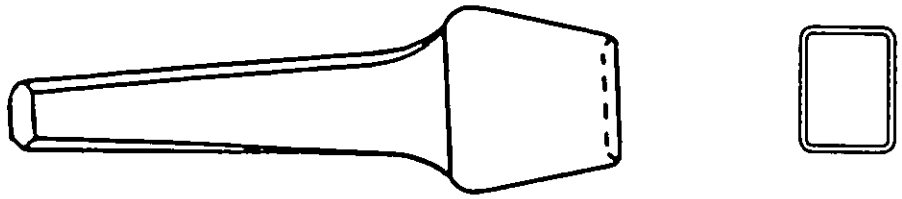


Figure 1. Type I, class A, cutting (drive) punch, rectangular (bag).

3.6.4 Class B, circular punch. The class B punch shall have a circular shaped cutting edge and shall be forged of tool or alloy steel. The design of the class B punch shall be such that the user may easily see the template or layout lines when the punch is positioned over the work for use.

3.6.4.1 Class B, style 1, double bow. The class B, style 1 punch shall be of the double bow, arch type, and shall be either forged in one piece or a steel shank shall be welded to the punch end at the point where cuttings are discharged. The style 1 punch shall be capable of punching holes conforming to the sizes shown in table II and be similar to figure 2.

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TABLE II. Type I, class B, style 1, cutting (drive) punch, circular, double bow

Size number	Diameter of hole	Length, overall ± 15 percent	Size number	Diameter of hole	Length, overall ± 15 percent
	Inches	Inches		Inches	Inches
3	3/8	5	17	1-3/8	6-1/2
4	7/16	5	18	1-1/2	6-1/2
5	1/2	5	19	1-5/8	6-1/2
6	9/16	5	20	1-3/4	7
7	5/8	5	21	1-7/8	7
8	11/16	5	22	2	7
9	3/4	5	23	2-1/8	7
10	13/16	5-1/2	24	2-1/4	7
11	7/8	5-1/2	25	2-3/8	7-1/4
12	15/16	5-1/2	26	2-1/2	7-1/4
13	1	5-1/2	27	2-3/4	7-3/4
14	1-1/16	6	28	3	7-3/4
15	1-1/8	6	29	3-1/4	8
16	1-1/4	6	30	4	8

Tolerances ± 0.010 for sizes up to 1 inch; ± 0.020 for sizes over 1 inch and up to 2 inches; ± 0.030 for sizes over 2 inches and up to 3 inches; and ± 0.040 for sizes over 3 inches and up to 4 inches.

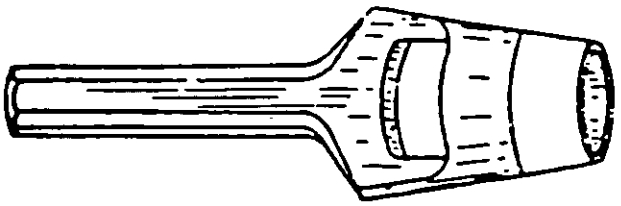


Figure 2. Type I, class B, style 1, cutting (drive) punch, circular, double bow.

3.6.4.2 Class B, style 2, single bow. The class B, style 2 punch shall be of the single bow type, commonly known as belt or saddlers' punch. The punch shall be capable of punching holes conforming to the sizes shown in table III and be similar to figure 3.

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TABLE III. Type I, class B, style 2, cutting (drive) punch, circular, single bow

Size number	Diameter of hole	Length, overall ± 15 percent	Size number	Diameter of hole	Length, overall ± 15 percent
	Inch	Inches		Inch	Inches
1	1/16	3-1/2	11	7/32	4
2	5/64	3-1/2	13	1/4	4
4	7/64	3-1/2	15	9/32	4
5	1/8	3-3/4	16	5/16	4
6	9/64	3-3/4	18	23/64	4
7	5/32	3-3/4	19	13/32	4
8	11/64	3-3/4	20	3/32	3-3/4
9	3/16	4	21	11/32	4

Tolerances ± 0.010 for sizes up to 1 inch; ± 0.020 for sizes over 1 inch and up to 2 inches; ± 0.030 for sizes over 2 inches and up to 3 inches; and ± 0.040 for sizes over 3 inches and up to 4 inches.



Figure 3. Type I, class B, style 2, cutting (drive) punch, circular, single bow.

3.6.5 Class C, tinners'. The class C tinners' punch shall be for punching holes in soft sheet metal.

3.6.5.1 Class C, style 1, hollow. The class C, style 1 punch cutting portion shall be circular in shape. The shank shall be of sufficient size to form a rigid tool and shall be either circular, square, rectangular, hexagonal, or octagonal in cross section. The circular cutting portion for punches of hole diameters of 5/8 inch and larger shall form a shoulder at approximately one-fourth the length of the punch. The circular cutting portion of punches of hole diameters of 1/2 inch and less shall be approximately one-fourth of the length of the punch. The style 1 punch shall be capable of punching holes conforming to the sizes shown in table IV and similar to figure 4.

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TABLE IV. Type I, class C, style 1, cutting (drive) punch, tinners', hollow

Size number	Diameter of hole	Length, overall ± 15 percent	Size number	Diameter of hole	Length, overall ± 15 percent
	Inch	Inches		Inch	Inches
1	1/4	4	9	1-1/4	5-1/2
2	5/16	4	10	1-1/2	6
3	3/8	4	11	1-3/4	6
4	1/2	4	12	2	6
5	5/8	5	13	2-1/4	6-1/2
6	3/4	5	14	2-1/2	6-1/2
7	7/8	5	15	2-3/4	6-1/2
8	1	5	16	3	6-1/2

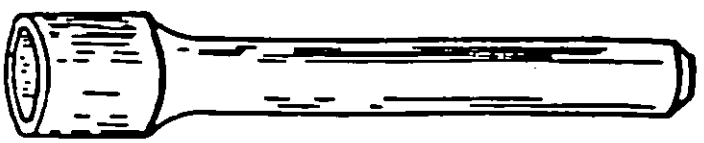


Figure 4. Type I, class C, style 1, cutting (drive) punch, tinners', hollow.

3.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.8 Workmanship. Details of workmanship shall be in accordance with the best commerical standards and practices. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for ensuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. The sampling procedures shall be in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. Visual examination of the end item shall be made for the defects set forth in table V. The lot size shall be expressed in terms of punches. The sample unit shall be one punch. The Inspection Level shall be II and the Acceptable Quality Level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements. The lot size shall be expressed in terms of punches. The sample unit shall be one punch. The Inspection Level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.3.3 Examination of containers and sets. Each sample unit shall be examined for any nonconformance with assembly of sets, including boxes or rolls (see 3.5 through 3.5.1.4). The sample unit shall be one punch. The Inspection Level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.5. The sample unit shall be one punch. The lot size shall be expressed in terms of punches. The Inspection Level shall be S-2 and the AQL, expressed as percent defective, shall be 2.5.

4.4.1 Performance test for cutting (drive) punch, class A and B, style 1 and 2. Each sample punch shall be tested by punching at least twenty holes in 1/8-inch-thick extra heavy leather, or flat, rubber conveyor belting in accordance with type A or MIL-B-52761. The leather shall be placed on a suitable hardwood backing. Attention shall be given to each sample punch as to whether the punchings pass through and are readily discharged from the barrel, where applicable, and as to whether the barrel, shank, and the striking end of the sample punch offer resistance to breakage, deformation, spalling, or cracking.

4.4.2 Performance test for cutting (drive) punch, class C, style 1. Each sample punch shall be tested by punching twenty-five holes in soft low carbon sheet steel with a thickness of approximately 0.026 inch. The cutting edge shall show ease in cutting and demonstrate the ability to hold a keen edge. Attention shall be given to each sample punch as to whether the striking end, shank, body, and taper offer proper resistance to breakage, deformation, spalling, or cracking.

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4.4.3 Hardness test. Each sample punch shall be tested in accordance with ASTM E18, ASTM E384, or a "file test" procedure.

4.4.3.1 When hardness is determined using a "file test" and the results are challenged or disputed, resolution shall be accomplished by testing in accordance with the procedures in ASTM E18 or ASTM E384, as applicable.

4.4.4 Flammability test for plastic wrap or roll. The plastic wrap or roll shall be rolled into a tight roll and clamped in a support, outward in a horizontal position. A 5 inch square piece of ten-mesh Bunsen burner gauze shall be clamped in a horizontal position, one-quarter inch below the bottom of the roll and 1/2 inch back from the free end of the roll. A Bunsen burner with an efficient flame 1/2 inch to 3/4 inch in height shall be adjusted so that the flame tip will contact the specimen when placed under the free end of the roll. At the end of thirty seconds, the flame shall be removed and the sample allowed to burn. The time required for the flame to travel from the end of the roll to the clamping device after the removal of the Bunsen burner shall determine the rate of travel of the flame. In case the roll does not continue to burn after the first ignition, the Bunsen burner shall be replaced under the free end of the roll for a period of thirty seconds immediately following the extinction of the flame on the roll. If the roll stops burning before the flame has reached the clamping device after the second ignition, the sample shall be considered self-extinguishing. When tested as specified, the plastic roll shall not burn at a rate exceeding 4 inches per minute.

4.4.5 Drop test. Drop testing shall be in accordance with the applicable paragraph(s) and table(s) of ANSI B 209.2. Drop testing for the 3/8-inch point diameter shall be performed as specified for the 1/4-inch point diameter.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. The lot size shall be the number of shipping containers in the end item inspection lot. The sample unit shall be one container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A or C, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class and style of punches required (see 1.2).
c. Size number of punches required (see applicable tables).
d. Whether punches in sets are required (see 3.5).
e. Whether sets should be furnished in wooden boxes, plastic boxes, fabric wraps or rolls, or plastic wraps or rolls (see 3.5.1).
f. Level of preservation, packaging, and packing required (see 5.1).

MILITARY INTERESTS:

DoD waives coordination

PREPARING ACTIVITY:

GSA - FSS

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